

Machine Learning

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Lecture 16: SoftMax

SoftMax

- Recall that the sigmoid produces a number between 0 and 1.
 - The sum of the probabilities of an email being either spam or not spam is 1.0.
 - But OvR, and OvO do not abide by these rules for the sums of probabilities across n classes
- SoftMax extends this idea into a multi-class world.
 - SoftMax assigns a probability to each class in a multi-class problem and the probabilities across classes add up to 1.0.
 - Works only when each example is a member of only one class (i.e., it does not work directly for multilabel)
 - If this is not true, rely on multiple logistic regression

A quick side note on math

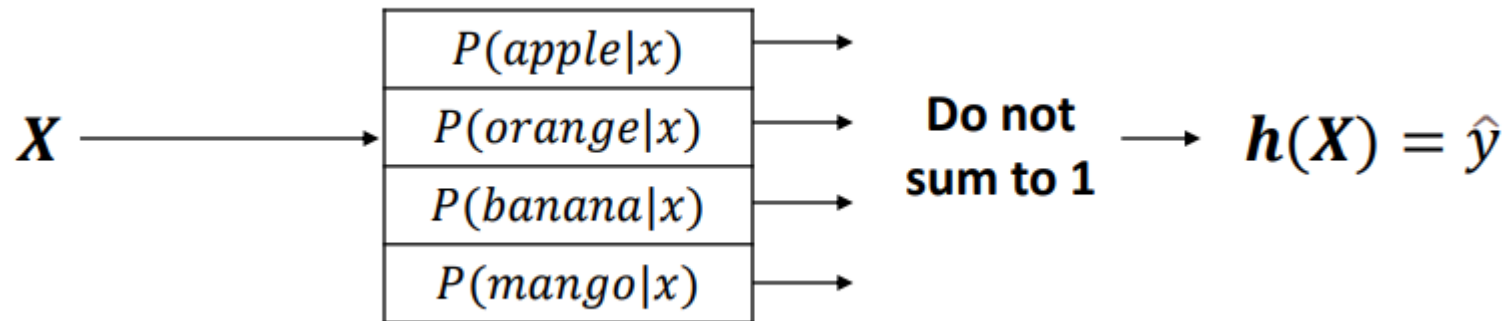
$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$= \frac{1}{1 + \frac{1}{e^x}} = \frac{1}{\frac{(e^x + 1)}{e^x}}$$

$$= \frac{e^x}{e^x + 1} = \frac{e^x}{e^x + e^0}$$

Generalizing Logistic Regression

- For multiple classes ($\#classes > 2$) we can generalize logistic regression to SoftMax regression
- One-of- C classes
- E.g. 4 classes (0, 1, 2, 3)



- SoftMax converts these C raw scores to probabilities that sum to 1.

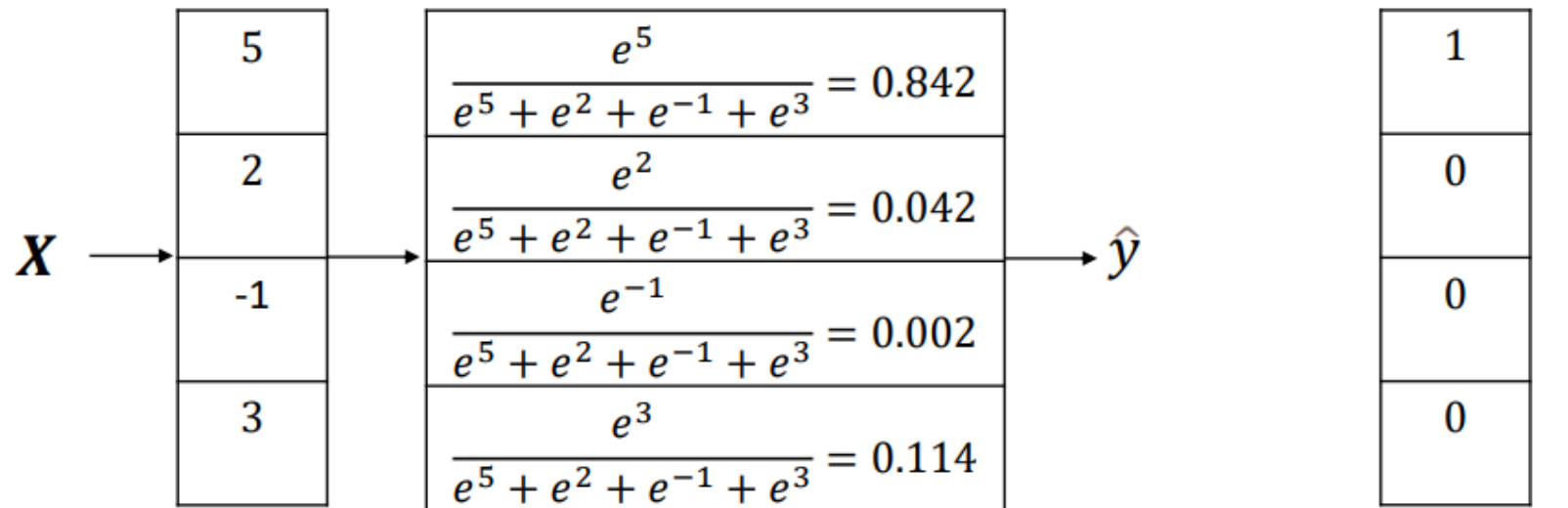
SoftMax activation function

$$z = wx + b$$
$$S(z)_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} = \frac{e^{z_i}}{e^{z_1} + e^{z_2} + \dots + e^{z_C}}$$

For 2 classes $\{z, 0\}$, the softmax is identical to sigmoid

$$S(z)_1 = \frac{e^z}{e^z + e^0} = \frac{e^z}{e^z + 1} = \sigma(z)$$

$$S(z)_2 = \frac{e^0}{e^z + e^0} = \frac{1}{e^z + 1} = 1 - \sigma(z)$$



Loss Function

$$y = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \hat{y} = \begin{bmatrix} 0.3 \\ 0.2 \\ 0.1 \\ 0.4 \end{bmatrix}$$

$$L(y, \hat{y}) = - \sum_{j=1}^c y_j \log \hat{y}_j$$

$$L(y, \hat{y}) = -(0. \log \hat{y}_1 + 1. \log \hat{y}_2 + 0. \log \hat{y}_3 + 0. \log \hat{y}_4)$$

$$L(y, \hat{y}) = -y_2 \log \hat{y}_2 = -\log \hat{y}_2$$

$$J(y, \hat{y}) = \frac{1}{m} \sum_{i=1}^m L(\mathbf{y}^{(i)}, \hat{\mathbf{y}}^{(i)})$$

Thank You